Finance tools, and the engineering deployment environment all used Cedar for policy evaluation, but each defined entities differently, cached signals on its own schedule, and decided independently which attributes to fetch and store.

This worked until it didn't. Policies that seemed similar on paper acted differently in practice because the context behind them wasn't consistent. For example, two systems might both try to express the same intent: allow employees who are currently active to access internal resources. In Cedar, that policy might look like this:

```
permit (
    principal,
    action,
    resource
)
when { principal.employmentStatus == "active" };
```

But not every system used the same vocabulary. In one PDP, HR data represented active employees as `active`, while another labeled them `enabled`. Even though the policy logic was identical, its evaluation could produce different results simply because the underlying entity data wasn't normalized. Debugging access decisions became tedious, and engineers spent more time reconciling data differences than writing policies.

To fix this, ACME's identity team began designing an enterprise Policy Information Point (PIP), a centralized signal normalization layer that would gather signals from all relevant systems, apply a common schema, and deliver normalized entities to every PDP.

12.6.1 The normalization layer

The normalization layer performs three main functions:

1. **Collect signals from various sources.** It connects with HR, Finance, device management, the customer information system, and external services like PagerDuty. Each connection specifies how signals are collected, how current they need to be, and how their authenticity and provenance are verified.